



## HACKSPHERE 2026 PHASE 1 PROPOSAL IDEA

| Project Title                                                |
|--------------------------------------------------------------|
| TANDATANI — A Verifiable Plot Passport for Every Smallholder |

| Team Identity |                             |
|---------------|-----------------------------|
| Team Name     | ROOT ACCESS                 |
| Team Category | International (All Members) |

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|-------------------------------|------------------------|
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| Student Status (Current)      | College Student        |
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| Institution                   | ⟨member 2 institution⟩ |

| Member 2 Personal Information |                        |
|-------------------------------|------------------------|
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| Institution                   | ⟨member 3 institution⟩ |



# 1. Executive Summary

**TANDATANI** (from Bahasa Indonesia *tanda* “mark” + *tani* “farmer” — an echo of *tanda tangan*, a signature) gives every smallholder plot a **cryptographically verifiable Plot Passport that the farmer owns**, so Indonesia’s 1.6 million coffee-farming families<sup>1</sup> — and after them the palm, cocoa and rubber smallholders behind up to **US\$7 billion** of exports<sup>2</sup> — are not locked out of Europe when the EU Deforestation Regulation takes effect on **30 December 2026**.<sup>3,4</sup> The registration that doubles as a farmer’s geolocation record, e-STDB, covers just **0.03% of coffee farmers** today;<sup>5</sup> TandaTani closes that gap at the price of a WhatsApp message.

A cooperative officer maps a plot in minutes on a **Bahasa-first, offline-capable app**; an **AI engine** cross-examines it against official satellite forest data (FAO Whisp, JRC-2020 baseline, GLAD/RADD alerts)<sup>6,7</sup> and explains its verdict in plain Bahasa; the cooperative then signs a **W3C Verifiable Credential**<sup>8</sup> the farmer keeps and can carry to *any* buyer, its evidence hash **anchored on the Polygon public chain** ( $\approx$  \$0.001 per thousand plots<sup>9</sup>). Exporters click once for an **EU-ready due-diligence dossier** (TRACES GeoJSON)<sup>10</sup> — and **anyone, including the judges reading this, can scan a QR and verify the entire chain of proof with no account, no permission, and no trust in us**.

Farmers pay nothing; exporters pay per shipment — less than 6% of a single consultant-day at today’s €800–1,500 rates<sup>11</sup> — for compliance that carries a documented **+US\$50/tonne premium** (Vietnamese-robusta benchmark).<sup>12</sup> Hybrid Web2 + Web3 + AI, every component free-tier, and every feature scoped to run **live, end-to-end, within the 24-hour final**.

## 2. Problem Statement

**The problem in one sentence:** from 30 December 2026, every kilogram of coffee, palm oil, cocoa or rubber entering the EU must carry plot-level geolocation proof that it grew on land that was not deforested after 2020 — and the Indonesian smallholders who grow almost all of it have effectively none of the required records.

### 2.1 A regulatory wall, eleven weeks after this hackathon’s final

Regulation (EU) 2023/1115 (EUDR) obliges any operator placing seven commodities on the EU market to file a due-diligence statement containing the **geolocation of every plot of origin** — a GPS point for plots up to 4 ha, a full polygon above that — plus evidence of **no deforestation after 31 Dec 2020** and of **legality** under Indonesian law.<sup>3,10</sup> After two postponements, enforcement begins **30 December 2026** for large and medium operators (30 June 2027 for micro/small);<sup>4</sup> the runway has been consumed, and the simplifications that came with the delay did not remove the plot-level data requirement. Non-compliance risks fines whose ceiling is no lower than **4% of a company’s EU-wide turnover**, plus confiscation and exclusion from public procurement.<sup>3</sup> Indonesia is benchmarked “**standard risk**” — 3% of operators face ex-post audits, but a valid DDS reference is a customs precondition for *every* shipment, and buyers facing the fines push the data burden upstream regardless.<sup>13</sup>

## 2.2 The people on the wrong side of the wall

|                                                                                                           |                                                                                                                                    |                                                                                                   |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>99.3%</b><br>of Indonesian coffee production comes from smallholders on 0.5–2 ha plots <sup>1,14</sup> | <b>0.03%</b><br>of coffee farmers hold e-STDB — the registration that doubles as their geolocation record (MoA, 2024) <sup>5</sup> | <b>US\$7 B</b><br>of Indonesian exports exposed to EUDR requirements (CMEA estimate) <sup>2</sup> |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|

Indonesia is the world's 4th-largest coffee producer (813,345 t in 2024)<sup>15</sup>, and its coffee is a smallholder crop to a degree few realize: **1.63 million farming households** work 98.1% of the coffee area.<sup>1</sup> The pattern repeats across the EUDR basket: smallholders hold **40.6% of the 16.8 M ha of oil palm** (≈2.4 M households),<sup>16,17</sup> grow **99.8% of cocoa**<sup>18</sup> and **≈91% of natural rubber**.<sup>19</sup> Yet the state instrument that would prove their plots' location and legality — the e-STDB cultivation registration — has reached **1.07% of palm farmers, 0.03% of rubber and coffee farmers, and 0.00% of cocoa farmers** (Ministry of Agriculture data, July 2024 — the latest published breakdown).<sup>5</sup> By December 2023 exactly **15,054** e-STDB certificates existed against ≈2.4 million oil-palm households alone.<sup>17</sup> Koltiva — the largest traceability provider in the country — itself estimates that **only ≈1% of Indonesian smallholders meet EUDR traceability and legality requirements**.<sup>20</sup>

### Field reality, August 2026

*In Pasaman, West Sumatra, Minangkabau families grow coffee inside ancestral parak agroforests — shade trees, areca, sugar-palm and coffee layered “like a tidied forest.” It is the most forest-friendly coffee agriculture imaginable. But the plots “have no polygons and are not recorded in any database” — so, come 30 December, the most forest-friendly coffee has the highest chance of being refused at Europe's door.<sup>21</sup> Meanwhile in Gayo Lues, cooperative staff are walking member plots one by one, mapping by hand against protected-area boundaries.<sup>22</sup>*

## 2.3 Why the gap persists — it is not a mapping problem

A GPS point takes thirty seconds to capture. The real bottlenecks, documented across government and field studies, are structural:

- **Cost asymmetry.** Traditional certification routes cost **€86/ha upfront and €191–751 per farmer**, and can push a smallholder's first-year net income **negative (–8%)**<sup>23</sup> — against typical plots of under 2 ha. Consultants bill **€800–1,500/day**.<sup>11</sup> These unit economics can never reach a 1-hectare farm.
- **Institutional throughput.** e-STDB issuance runs through district offices with limited capacity; the Agriculture Ministry has now opened data collection to third parties precisely because the state cannot map millions of plots alone.<sup>24</sup>
- **A trust and custody gap.** Records collected by enterprise platforms belong to the paying client, not the farmer; Deputy Foreign Minister Havas Oegroseno put the farmer's side bluntly — asked whether smallholders can perform EUDR's screening duties: **“Can farmers do it? Of course not.”**<sup>25</sup>



- **The digital divide is real but navigable.** Only **46.8%** of Indonesia's 28.2 M farmers use modern machinery or digital technology;<sup>26</sup> any workable solution must run offline, in Bahasa, on cheap Android phones — or through a literate cooperative officer.
- **Chain-of-custody fraud.** Collection points mix compliant and non-compliant lots; without tamper-evident records, one launderer can contaminate — legally and reputationally — an entire container.<sup>27</sup>

## 2.4 The cost of doing nothing

The market is already pricing the wall in. EU-bound Indonesian coffee shipments jumped **+72% to 2.4 million bags** in 2025/26 as buyers front-load stock ahead of enforcement<sup>28</sup> — stockpiling, not confidence. Compliant robusta already earns a **+US\$50/tonne premium** in Vietnam, where 35–40% of area is EUDR-ready; non-compliant beans are not discounted into Europe — they are **excluded**, because mixing contaminates the lot.<sup>12,27</sup> Physical segregation adds up to **+25%** logistics cost.<sup>27</sup> At stake for Indonesia: up to **US\$7 B of exports** by the government's own estimate,<sup>2</sup> at the very moment the newly concluded EU–Indonesia CEPA (Sept 2025) finally opens the €28.9 B trade corridor wider.<sup>29</sup> The door to Europe is being unlocked and walled up in the same year — and whether a farming family stays inside is decided by data they do not have.

**Synthesis.** Indonesia does not have a deforestation-evidence problem so much as a **proof-distribution** problem: the forests were standing in 2020, the satellites saw them, the farmers were there — but no instrument turns that truth into a portable, trusted, affordable record in a smallholder's hand. That is precisely an infrastructure gap — and infrastructure is what we build.

# 3. Proposed Solution & Value Proposition

Indonesia's harvest is real, the land is clean, but the evidence lives in paper folders, WhatsApp photos and enterprise databases the farmer will never control. TandaTani's answer: give the proof itself to the person who created it. Every ingredient we use exists; the **combination** — a compliance credential the *farmer* owns, priced *per shipment*, verifiable by *anyone* without an account — exists nowhere in production, and §3.5 shows why incumbents cannot copy it without breaking their own business model.

## 3.1 The core concept — a Plot Passport, owned by the farmer

For every plot we assemble one **evidence bundle**: geolocation (a GPS point for plots  $\leq 4$  ha, a walked polygon above — exactly the EU specification<sup>10</sup>), the satellite deforestation verdict, legality documents, geotagged field photos and the review trail. After human review, the cooperative signs this bundle as a **W3C Verifiable Credential 2.0**<sup>8</sup> — the *Plot Passport*. Three properties make it more than a database row:

- **Portable.** The passport is a signed credential held for the farmer — not locked in our database. If the farmer changes buyer or platform, or if TandaTani itself disappears, the proof stays valid and verifiable. No incumbent offers this; their records live in the paying client's silo.<sup>30</sup>
- **Independently verifiable.** Each bundle's SHA-256 hash is batched into a Merkle tree whose root is anchored on the Polygon public chain.<sup>9</sup> Any importer or auditor can re-compute the hash in a browser and confirm the record existed, unaltered, at issuance — without asking us, or trusting us.



- **Private by design.** The chain carries only salted hashes — never coordinates, never names. Geolocation is disclosed solely inside the exporter’s due-diligence submission, as the regulation requires.<sup>3</sup>

**Scope honesty — the legality pillar.** EUDR demands two proofs: deforestation-free *and* legal production. TandaTani verifies the first from satellites; for the second it structures and hash-seals the farmer’s legality evidence (e-STDB, land letters, cooperative membership), checks it for presence and internal consistency, and records who attested it — while authenticity verification remains with the issuing authorities. The dossier is evidence the exporter files under its own due-diligence responsibility; we do not replace that duty — we make it affordable to discharge.

### 3.2 How it works — one journey

Ibu Sari, a field officer of a Gayo cooperative, visits member farms with a mid-range Android. At Pak Karim’s plot she opens **Ladang**, our farmer app: one tap for the GPS point, two photos, a harvest estimate — all offline; it syncs at the next signal. Overnight, the **AI engine** interrogates FAO Whisp and Global Forest Watch: was this forest on 31 Dec 2020? Any loss alerts since?<sup>6,7</sup> PostGIS checks no other member has claimed an overlapping polygon. The verdict returns in Bahasa: “*Lahan ini aman — tidak ada deforestasi terdeteksi setelah 2020,*” with every evidence layer attached. The cooperative chair reviews her queue, taps **Terbitkan** (“issue”), and the cooperative’s key signs Pak Karim’s Plot Passport. When an exporter builds its next container for Hamburg, it selects verified passports and TandaTani emits the TRACES-format GeoJSON dossier<sup>10</sup> plus a QR-coded evidence certificate. The importer scans the QR: hash matches, signature valid, root on-chain — due diligence the importer can re-verify, byte for byte, five years from now.

### 3.3 Why Web3 — an honest answer, not a buzzword

- **The trust gap is structural.** An EU importer must rely on evidence from a smallholder they will never meet, passed through 3–5 intermediaries — some of whom profit from laundering non-compliant volume into compliant lots.<sup>27</sup> A vendor’s private database asks the importer to trust the vendor and every editor; a public anchor makes tampering *detectable by anyone* — a signature proves *who* wrote a record, the anchor proves *when*, so no issuer can quietly backdate a passport after a deforestation alert lands.
- **Records must outlive companies.** EUDR requires operators to keep due-diligence records for **five years**.<sup>3</sup> Startups die; anchored hashes on a public chain do not.
- **Portability breaks lock-in.** Verifiable Credentials give farmers *exit rights* over their own compliance history — the structural opposite of enterprise traceability, and the reason a farmer-first platform can win cooperatives’ participation.
- **Decentralized authentication where it belongs.** Exporters sign in with their wallets (SIWE); cooperatives hold signing keys (DIDs); farmers never touch crypto — a relayer pays the sub-cent gas. Web3 machinery, zero Web3 friction.

### 3.4 Why AI — verification at a price a smallholder can afford

Manual verification costs €800–1,500 per consultant-day;<sup>11</sup> corporate mapping programs have burned millions. TandaTani’s engine performs the equivalent convergence-of-evidence analysis in seconds for **≈ \$0.002 per plot**: deterministic geospatial scoring over official datasets (JRC forest baseline 2020, GLAD/RADD alerts, commodity maps via Whisp, with Global Forest Watch as an independent backup track)<sup>6,7</sup> plus PostGIS geometry-anomaly detection against duplicate or overlapping claims — then **Gemini** (official Bahasa support<sup>31</sup>)



renders the verdict as three plain sentences a farmer can act on, and powers **Tanya Tani**, a field-question copilot. The design principle is auditable honesty: **AI narrates; evidence decides**. Every verdict links its raw sources.

### 3.5 The landscape — and the wedge it leaves open

| Today's options                                                                 | Who pays, roughly                                            | The gap TandaTani exploits                                                                                                          |
|---------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Enterprise traceability — Koltiva, osapiens, Sourcemap, Agridence <sup>30</sup> | Exporters/brands: €10k–50k+/yr + €5k–90k setup <sup>11</sup> | Top-down; the farmer is a data row; records sit in the client's silo                                                                |
| Web3 agri-platforms — Dimitra × PT Surveyor Indonesia <sup>32</sup>             | Enterprise deals; fees tied to the DMTR token                | Token volatility as a compliance dependency; credentials not farmer-owned                                                           |
| NGO transparency — Fairfood Trace (Hedera) <sup>33</sup>                        | Donor-funded pilots                                          | Premium storytelling, not EUDR dossier automation; no AI verification                                                               |
| Consultants & certifiers <sup>11,23</sup>                                       | €800–1,500/day; €191–751 per certified farmer                | Unit economics that can never reach a 2-ha farm                                                                                     |
| Free public tools — Whisp, GFW, TRACES, Fairtrade Plot Insights <sup>6,34</sup> | Free                                                         | Analysis and filing exist — identity, attestations, volume-bounded dossiers, offline Bahasa capture and portable credentials do not |

**Unique selling proposition.** TandaTani is the only design in this landscape where the *farmer* ends up owning a portable, publicly verifiable compliance credential — free — while the exporter pays per shipment instead of per year. Incumbents cannot copy this without dismantling their enterprise pricing and data custody; that structural conflict, not secrecy, is our moat. And we are deliberately **complementary to the state**: Indonesia's blockchain-based National Dashboard and the e-STDB drive need exactly the verified, plot-level smallholder data we produce — the Agriculture Ministry now invites third parties to collect it.<sup>24,35</sup> TandaTani is the missing last mile between a farmer's field and every rail built above it.

## 4. Minimum Viable Product (MVP) / Key Features

Everything below will be **built and running end-to-end during the 24-hour final** — live backend, live satellite queries, live chain writes, on a public URL. Nothing hardcoded, nothing mocked. Scope discipline is the strategy: five features, each independently demoable, composing one continuous story.

### Feature 1 — Ladang: offline-first plot capture (Bahasa)

A PWA for low-end Android. One-tap GPS point for plots  $\leq 4$  ha (per the EU geolocation spec;<sup>10</sup> smartphone GPS reaches  $\leq 4.9$  m in open sky<sup>36</sup>, and under shade canopy Ladang shows live accuracy, accepting a point only below a 10 m threshold with tap-on-imagery fallback) or walk-the-boundary polygon capture above 4 ha, over Esri satellite imagery (MapLibre GL). Farmer profile, plot photos, harvest estimate. Captures queue in IndexedDB and sync when signal returns — demonstrated live in airplane mode.

### Feature 2 — AI verification engine

On sync, the backend submits each geometry to **FAO Whisp**<sup>6</sup> (convergence of evidence across JRC forest-cover 2020, tropical-moist-forest, GLAD-L/S2 and RADD alerts, ESA WorldCover, commodity maps) with the **Global Forest Watch API** as an independent



second track.<sup>7</sup> PostGIS runs ST\_Intersects duplicate/overlap detection against every existing plot. Results fuse into a deterministic risk score; **Gemini 2.5 Flash**<sup>31</sup> renders the verdict as three plain-Bahasa sentences with evidence links. The demo shows a clean plot passing and a test polygon drawn over post-2020 forest loss failing — live.

**Feature 3 — Plot Passport issuance (Cooperative Console)**

A review queue for the cooperative: map, AI verdict, photos and anomaly flags side by side; approve or reject in one tap. On approval the platform issues a **W3C Verifiable Credential 2.0**, Ed25519-signed by the cooperative’s did:key.<sup>8</sup> The passport — with its QR — appears instantly in the farmer’s app.

**Feature 4 — Public anchoring & one-scan verification**

Evidence hashes batch into a Merkle tree; the root is anchored by **TandaRegistry.sol** on Polygon (Amoy testnet in the demo; the identical call costs ≈ \$0.001 on mainnet<sup>9</sup>). The public verify page re-computes the bundle hash in-browser and checks the VC signature plus the Merkle proof against the on-chain root — no login, no API key, no trust in TandaTani.

**Judges can scan the QR from their own phones during the pitch.**

**Feature 5 — Exporter DDS Studio**

Wallet sign-in (SIWE); pick verified passports into a batch; export a due-diligence dossier — **TRACES-conformant GeoJSON** (WGS-84, six-decimal precision, point-or-polygon by plot size<sup>10</sup>) plus a QR-stamped evidence-summary PDF. A **volume-plausibility gate** blocks any dossier whose tonnage exceeds the summed harvest estimates of its passports — the launderer’s math stops working — and an append-only audit log shows every state change behind it.

**The 24-hour execution plan (3 builders × 24 h)**

| Window (WIB) | Builder 1 (S. Gupta) — Web3 / Infra                                                             | Builder 2 — Frontend / PWA            | Builder 3 — Backend / AI                |
|--------------|-------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------|
| 10:00–14:00  | Repo/CI; Vercel + Supabase; deploy TandaRegistry.sol to Amoy                                    | App shell; map capture; offline queue | Supabase schema + PostGIS; API skeleton |
| 14:00–19:00  | VC issuance service (did:key, Ed25519)                                                          | Farmer flows; verification result UI  | Whisp + GFW integration; risk scorer    |
| 19:00–00:00  | Merkle batcher + anchoring worker                                                               | Cooperative Console                   | Gemini Bahasa verdicts; Tanya Tani      |
| 00:00–05:00  | Public verify page (hash + proof in browser)                                                    | Exporter DDS Studio                   | TRACES GeoJSON + PDF dossier generator  |
| 05:00–08:00  | Integration pass, all hands — seeded Gayo demo dataset, airplane-mode drill, cross-device tests |                                       |                                         |
| 08:00–10:00  | Hardening, deploy freeze, pitch rehearsal ×3, Q&A drill                                         |                                       |                                         |

Feasibility rests on rehearsed pieces, not hope. Every external service (Whisp, GFW, Gemini, Amoy faucet, Supabase, Vercel) is free-tier, with accounts and API keys provisioned before the clock starts. Fallbacks are pre-planned: GFW if Whisp queues slowly; Groq/Llama behind the same interface if Gemini rate-limits; a cached verification fixture exists *only* as an offline-resilience aid and would be labeled as such on screen, never presented as live output. **Cut-not-fake rule: if a feature slips, we cut it — we never simulate it.** Declared cut order if time slips: the Tanya Tani copilot goes first, the PDF dossier second (the TRACES GeoJSON

remains), Exporter Studio polish third — the capture → verify → issue → publicly-verify loop (Features 1–4) is the non-negotiable demo core.

**Deliberately out of the 24-hour scope** — native mobile builds, payments, marketplace, mainnet deployment, SD-JWT selective disclosure, voice interface. They appear in §5 as roadmap and are promised nowhere as demo.

## 5. Target Market & Impact Viability

### 5.1 Exactly who uses TandaTani

- **Primary — cooperative field officers and managers** in Indonesia’s Arabica belt: specifically the Gayo highlands (Aceh Tengah, Bener Meriah, Gayo Lues), home to Southeast Asia’s largest Arabica area, a protected Geographical Indication origin since 2010, and cooperatives like women-led **Kopepi Ketiar** (2,000+ members across 19 villages) — which face a **January 2027** Fairtrade geolocation deadline on top of EUDR.<sup>34,37</sup> These are digitally literate professionals mapping member plots *by hand* today.<sup>22</sup>
- **Primary — compliance managers at Indonesian coffee exporters** shipping green beans to Belgium and Germany (the top two EU destinations<sup>28</sup>), who currently face consultant day-rates or enterprise SaaS to produce every due-diligence statement.<sup>11</sup>
- **Secondary (year 1–2)** — cocoa and rubber cooperatives (0.00% and 0.03% e-STDB coverage<sup>5</sup>), then independent palm smallholder associations; **EU importers** who verify passports (free) and later pay for API access.
- **Validation, honestly stated:** we have not yet interviewed a customer. Between this submission and the final we will hold structured calls with Gayo cooperative officers (intro path: the WRI Sustainable Gayo Coffee network and Fairtrade producer support) to test willingness-to-pay — and we will report what we hear, including bad news, on stage.

### 5.2 Market size, from the bottom up

| Scope                                      | Basis                                                                                                                                          | Size                         |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| TAM — EUDR due-diligence software, global  | Analyst estimate, 2026 → 2036 <sup>38</sup>                                                                                                    | US\$1.1 B → 1.8 B            |
| SAM — Indonesian smallholder EUDR corridor | ≈7.5 M smallholder households across coffee (1.63 M), palm (2.4 M), cocoa (1.4 M), rubber (2.1 M); exports at stake up to US\$7 B <sup>2</sup> | ≈7.5 M farms · US\$7 B trade |
| SOM — 12-month beachhead: Gayo coffee      | 3 co-ops · 8,000 members × €1.40 × 2 harvest seasons + 2 exporters · 200 dossiers × €45 ≈ €31k                                                 | ≈US\$33k ARR                 |

### 5.3 Business model — commercial from day one

- **Farmers: free, forever.** Their participation is the network; charging them kills it.
- **Exporters: Rp 800k (≈ €45) per due-diligence dossier**, or a flat coop-season license. Context: consultants run €800–1,500/day, enterprise SaaS €10k–50k/yr;<sup>11</sup> one compliant 19-tonne container carries ≈ US\$950 of premium at +US\$50/tonne.<sup>12</sup> We price under 5% of the value we unlock.

- **Cooperatives: Rp 25k (≈ €1.40) per member per season** for the credential registry and Fairtrade/e-STDB exports — less than 1% of a certification’s per-farmer cost;<sup>23</sup> a member visit takes minutes, and one visit now serves EUDR, Fairtrade and e-STDB at once.
- **Later: verification API** for EU importers and (per Ditjenbun’s invitation to third parties<sup>24</sup>) **paid e-STDB data-collection partnerships** feeding the National Dashboard.<sup>35</sup>

**Unit economics.** Marginal cost per verified plot ≈ \$0.002 LLM + anchoring amortized to ≈\$0.000001 in 1,000-plot batches + fractions of a cent of storage<sup>9,31</sup> — under **\$0.01 all-in** against €45-per-dossier revenue: software margins on infrastructure that incumbents deliver with field armies. Free tiers carry the pilot; the first paying exporter carries the company.

## 5.4 Scalability — one engine, many walls

The pipeline — *map* → *verify* → *credential* → *anchor* → *dossier* — is commodity-agnostic and regulation-agnostic. The same engine serves cocoa, rubber and palm; the same passports satisfy Fairtrade’s 2027 geodata mandate,<sup>34</sup> feed e-STDB and the National Dashboard,<sup>35</sup> and are ready for whatever follows EUDR (the UK and US have parallel bills in motion). Geographically, Vietnam alone has 60–65% of its coffee area still non-compliant;<sup>12</sup> the architecture ports with a language file. Because passports are portable open credentials, every new buyer that accepts them makes every existing passport more valuable — adoption compounds.

## 5.5 Impact, measured

- **Economic:** keeping smallholders inside a corridor worth up to US\$7 B<sup>2</sup> and capturing the +US\$50/t compliance premium<sup>12</sup> at the farm gate rather than the trader’s margin.
- **Institutional:** every passport is e-STDB-ready data — accelerating the registration drive the government itself says it cannot staff.<sup>24</sup>
- **Environmental:** EUDR’s promise only works if compliance is achievable; TandaTani makes deforestation-free farming *provable*, so the market can reward it (SDGs 1, 8, 12, 13, 15).
- **Social:** the first bankable, portable proof-of-practice a smallholder has ever owned — with women-led cooperatives like Ketiaara<sup>37</sup> as first protagonists.

## 5.6 Risks, named — and answered

| Risk                                                                | Reality check                             | Mitigation built into the design                                                                                                                                                      |
|---------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Farmer digital literacy (46.8% modern-tech adoption <sup>26</sup> ) | Real, generational                        | Cooperative-officer-assisted capture; offline PWA; Bahasa-first; voice roadmap                                                                                                        |
| EUDR slips a third time                                             | Possible — it slipped twice <sup>4</sup>  | The 2024 delay itself triggered +72% buyer front-loading — buyers, not Brussels, now enforce; <sup>28</sup> passports also serve Fairtrade Jan-2027, e-STDB and ISPO <sup>24,34</sup> |
| GPS spoofing / claim fraud                                          | Attempted everywhere                      | Satellite cross-check, PostGIS overlap detection, geotagged photos, cooperative attestation — convergence, not trust                                                                  |
| Cooperative key loss                                                | Operational hazard                        | Credential-status registry + re-issuance under a successor DID (did:key is single-keypair by design; roadmap: did:web for true rotation)                                              |
| Incumbent copies the model                                          | Koltiva has 1.9 M producers <sup>30</sup> | Copying farmer-owned portability breaks their enterprise custody model — our wedge is their conflict                                                                                  |

## 6. Tech Stack & System Architecture

One hybrid system, three planes: a **Web2 core** for speed and cost, a **Web3 trust plane** for proof and portability, an **AI plane** for verification at smallholder prices. Every component is free-tier; the whole demo runs on public URLs.

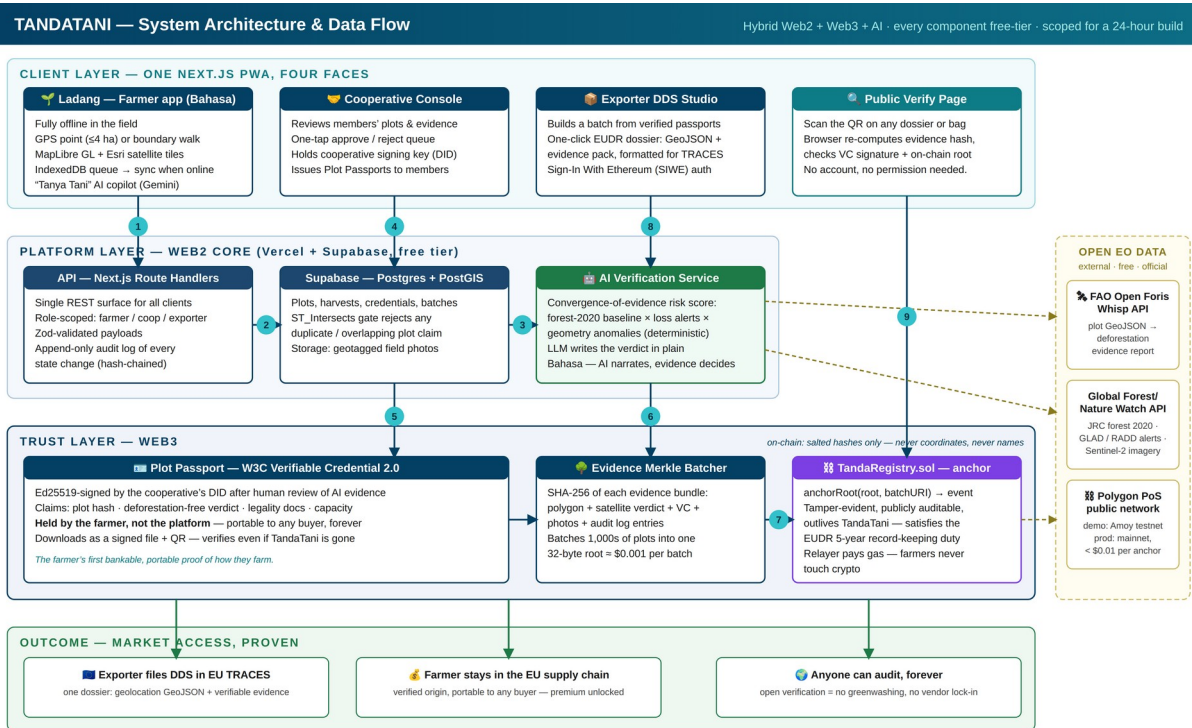


Figure 1 — TandaTani system architecture and data flow. Numbered chips on the arrows match the data-flow steps in §6.1.

### 6.1 Data flow (follows the diagram's numbering)

1. Field capture, fully offline: GPS point/polygon, photos, farmer identity → IndexedDB queue.
2. Sync to the API (Next.js route handlers, Zod-validated); plots land in Postgres/PostGIS,<sup>39</sup> where the ST\_Intersects integrity gate rejects duplicate and overlapping claims on arrival.
3. AI engine gathers satellite evidence (Whisp primary; GFW independent track)<sup>6,7</sup> and Gemini writes the Bahasa verdict.<sup>31</sup>
4. Cooperative reviews the AI-annotated queue in its Console.
5. Approved bundle → W3C Verifiable Credential 2.0, signed with the cooperative's did:key (Ed25519).<sup>8</sup>
6. Bundle hashes batch into a Merkle tree — thousands of plots, one 32-byte root.
7. The root is anchored by TandaRegistry.sol on Polygon; the event is public forever.<sup>9</sup>
8. Exporter composes verified passports → TRACES GeoJSON + evidence-PDF dossier.<sup>10</sup>
9. Anyone scans the QR: the browser re-computes the hash and checks VC signature + Merkle proof against the chain.

### 6.2 Technology choices — and why each earns its place

| Layer | Technology | Why this, for this build |
|-------|------------|--------------------------|
|-------|------------|--------------------------|



|          |                                                                                                                                   |                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Frontend | Next.js 15 PWA · TypeScript · Tailwind · MapLibre GL + Esri World Imagery · Dexie (IndexedDB)                                     | One codebase, four roles; installable on low-end Android; offline by architecture, not afterthought             |
| Backend  | Next.js route handlers on Vercel · Supabase: Postgres + PostGIS, Auth, Storage <sup>39</sup>                                      | Zero-ops free tiers; geometry fraud checks are one SQL clause                                                   |
| AI / EO  | FAO Whisp API · Global Forest Watch Data API · Gemini 2.5 Flash, Groq/Llama fallback <sup>6,7,31</sup>                            | Official, free, EUDR-aligned convergence of evidence; Bahasa officially supported                               |
| Web3     | Solidity (Foundry) on Polygon PoS/Amoy · viem · @digitalbazaar/vc + did:key (Ed25519) · SIWE · OpenZeppelin Merkle <sup>8,9</sup> | Sub-cent anchoring; W3C-standard credentials (Recommendation, May 2025); wallets only where users are companies |
| Delivery | GitHub Actions CI · Vercel previews · public demo URL + Amoy explorer links                                                       | Live public deployment for the pitch — judged explicitly above localhost                                        |

### 6.3 The on-chain / off-chain boundary (privacy by architecture)

- **On-chain:** 32-byte Merkle roots and batch URIs — nothing else.  $\approx$  \$0.001 per thousand plots<sup>9</sup>: the whole Gayo highlands could be anchored for the price of one cup of its own coffee.
- **Off-chain:** coordinates, identities, photos — Postgres under row-level security, disclosed only inside the exporter's DDS as EUDR requires.<sup>3</sup> Salted hashing defeats rainbow-matching of known plots.
- **Keys:** cooperatives hold did:key signing keys; exporters authenticate by wallet (SIWE); a relayer pays gas so farmers never see crypto.
- **Privacy law, named.** Indonesia's PDP Law (UU 27/2022) and the GDPR shape the design: consent captured at enrollment (the cooperative as data controller, TandaTani as processor), coordinates under row-level security and disclosed only inside the DDS, credential copies on the farmer's device and printable QR card — and, because the chain holds only salted hashes, destroying a record's salt is a clean erasure path. The public QR page shows a verdict, never coordinates or identities.

### 6.4 Engineering honesty — failure modes we designed for

- Whisp jobs are asynchronous (Earth-Engine-backed): we submit early, poll, and keep GFW as an independent evidence track — either alone yields a verdict.<sup>6,7</sup>
- Background Sync API is Chromium-only, so the offline queue flushes on the browser's online event — works everywhere modern; demoed on Android Chrome.
- Free-tier LLM limits (Gemini: 250 requests/day<sup>31</sup>) comfortably exceed rehearsal-measured demo load; Groq is wired behind the same interface.
- If venue Wi-Fi dies mid-pitch, the offline-first design *is* the contingency: capture continues, and we sync over any hotspot.

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